Assessment: Technical Product Manager

COMPREHENSIVE SUBMISSION

Submission for FactWise | Candidate Name: TPM Assessment

Problem Statement & Operational Context

A specialized quarantine and treatment facility managing a high-consequence viral outbreak (15% mortality rate benchmark, 74-bed capacity at full occupancy) is suffering from critical operational bottlenecks:

- **Unsynchronized Rounds:** Doctors visit patients before nurses take temperatures, wasting clinical bandwidth.
- **Discharge Bottlenecks:** Patients stay an extra 1–2 days because 3-day fever-free recoveries are not promptly communicated to administration.
- **Zero Telemetry:** No visibility into treatment efficacy or monitoring against the 15% mortality / 85% cure benchmark.

Q1: User Types, Features & Action Mapping (MVP)

The solution structures workflows across 4 distinct user personas:

User Persona	Core Responsibility & Pain Point Solved	Features & Permitted Actions
1. Nurse	Daily vitals round accountability; eliminates journal logging, duplicate measurements, and missed checks.	• **Daily Queue Filter:** Toggle between *Needs Check* and *Completed Today*. • **Record Temperature Modal:** Input in °F with instant color feedback (Green: <99.5°, Orange: 99.5–100.3°, Red: ≥100.4°F). • **Streak Tracker:** Real-time visual progress counter toward the 3-day fever-free discharge rule. • *Restricted:* Cannot approve discharges or alter bed admissions.
2. Attending Doctor	Eliminates wasted trips prior to vitals; reviews trendlines; authorizes cures without administrative latency.	• **Prioritized Queue:** Automatically groups patients into *Ready for Round* (vitals done) vs *Pending Vitals* (warning). • **Vitals Sparkline:** 7-day temperature curve with a 100.4°F threshold marker. • **Clinical Notes:** Prescribe medications and document shift assessments. • **Discharge Gatekeeper:** 1-click discharge sign-off strictly unlocked once 3 consecutive fever-free days are verified.
3. Facility Admin	Strict 74-bed occupancy control; patient intake and discharge logistics; freeing beds immediately upon patient departure.	• **74-Bed Interactive Matrix:** Visual tiles 1–74 color-coded by real-time status (Red: Occupied, Amber: Discharge Pending, Green: Available). • **Discharge Desk:** Instant queue of doctor-approved discharges; 1-click "Finalize Discharge & Release Bed". • **Patient Intake Form:** Validates bed availability (strictly blocks admissions when active census = 74). • **Mortality Recording:** De-registration tracking outcome and timestamp.
4. Head Doctor / Medical Director	Clinical quality oversight; real-time comparison against the 15% mortality / 85% success benchmark.	• **Quality Scorecards:** Live Mortality Rate vs 15% threshold and Cure Rate vs 85% benchmark. • **Automated Alerts:** Critical warning banner triggered if mortality exceeds 15.0%. • **Shift Compliance Progress:** Real-time gauges tracking

User Persona	Core Responsibility & Pain Point Solved	Features & Permitted Actions
		Nurse Checks % and Doctor Rounds % for the day. • **Length of Stay (ALOS):** Rolling average stay duration to detect bed-blocking.

Q3: Logical Constraints & Technical Rules for Developers

In addition to UI screen designs, developers must enforce the following clinical and architectural constraints:

1. Sequential Rounding Dependency (Nurse-First Rule)

- Daily patient vitals flags (`todayTempRecorded` and `todayDoctorVisited`) reset to `false` at `00:00:00` daily.
- If an Attending Doctor opens a patient record marked *Pending Vitals*, the system displays an amber warning: *"Warning: Temperature has not been recorded today by a nurse."* The doctor is given the option to record the temperature directly, which tags the record as `recordedBy: "doctor"` and clears the nurse queue item.

2. The Strict 3-Day Fever-Free Cure Algorithm

- **Fever Definition:** Any reading $T \ge 100.4^\circ\text{F}$ (38.0°C) constitutes a fever.
- **Streak Reset:** If on Day 3 a patient logs 100.4°F or higher, the fever-free counter **strictly resets to Day 0**.
- **Discharge Permission Gate:** The "Sign Discharge Order" action is disabled in the API and UI unless `consecutiveFeverFreeDays ≥ 3`.
- **Two-Stage Handoff:** Doctor approval sets status to `doctor_approved_discharge` (alerting Admin desk). Only Admin completion sets status to `discharged` and frees the bed in the 74-bed matrix.

3. Capacity & Calculation Formulas

- **Capacity Ceiling:** Hard ceiling at 74 concurrent active patients. Attempts to admit when active census = 74 returns HTTP 422 with a descriptive error.
- **Closed Cases Formula:** To avoid distorting rates with active inpatients, mortality and cure rates are calculated exclusively on resolved cases: $$\text{Mortality Rate} = \frac{\text{Total Deceased}}{\text{Discharged Cured} + \text{Deceased}} \times 100, \quad \text{Cure Rate} = \frac{\text{Discharged Cured}}{\text{Discharged Cured} + \text{Deceased}} \times 100$$
- **Threshold Alert:** If Mortality Rate > 15.0%, trigger high-priority alert banner across all administrative views.

Q4: Sprint Planning Mathematical Optimization

Context: Team Velocity = 25 Story Points (SP). Already committed: 3 stories × 5 SP = 15 SP. Remaining capacity = **10 Story Points**.
Value Weights: Low = 1 unit ($V_L = 1$), Medium = 3 units ($V_M = 3$), High = 6 units ($V_H = 6$).

Feature	Priority	Value (V)	Effort (W)	Value Density (V/W)	Selection Status
J	Low	1	0.1	**10.00**	**SELECTED**
A	High	6	1.0	**6.00**	**SELECTED**
I	Low	1	0.3	**3.33**	**SELECTED**
F	Medium	3	1.0	**3.00**	**SELECTED**
G	Low	1	0.4	**2.50**	**SELECTED**
B	High	6	3.0	**2.00**	**SELECTED**
H	Low	1	0.5	**2.00**	**SELECTED**
D	Medium	3	2.0	**1.50**	**SELECTED**

Feature	Priority	Value (V)	Effort (W)	Value Density (V/W)	Selection Status
C	High	6	5.0	1.20	Excluded (Heavy effort: 5 SP for 6 V)
E	Medium	3	3.0	1.00	Excluded (Low density: 1.0)

Optimization Comparison

Strategy	Features Included	Total Effort	Total Value Delivered	Buffer Remaining
Standard High-First	A, B, C, F	10.0 SP	21 Value Units	0.0 SP (Zero tolerance)
Optimal Knapsack	**A, B, D, F, G, H, I, J**	**8.3 SP**	**22 Value Units**	**1.7 SP (Safety buffer)**

> **Final Mathematical Recommendation:** Select **{A, B, D, F, G, H, I, J}**. This combination mathematically maximizes value at **22 points** (exceeding all alternatives) while consuming only **8.3 story points**, providing an essential 1.7 SP safety buffer to protect team velocity.

Q2: Functional Prototype Implementation & Verification

A fully working web application has been constructed with Next.js, Tailwind CSS, TypeScript, and Recharts, and is ready for immediate deployment:

- **Workspace Directory:** `C:\Users\kaulj\.gemini\antigravity\scratch\quarantine-facility-portal`
- **Build Verification:** Compiled cleanly with zero errors (`next build` verified, static pages generated).
- **Git Version Control:** Initialized and committed (`main` branch, 15 files).

Key Demonstrable Workflows Built Into the Prototype:

1. **Nurse Workflow (Streak Progression):** Patients at Bed 5, 12, and 28 are pre-loaded with exactly 2 fever-free days. Recording a normal temperature ($<100.4^\circ\text{F}$) advances the counter to 3/3 and automatically changes status to *Eligible for Discharge*.
2. **Doctor Workflow (Clinical Review & Signoff):** Bed 40 has met the 3-day criterion. Opening Bed 40 displays the 7-day temperature trendline and activates the glowing green "Sign Discharge Order" button.
3. **Admin Workflow (Discharge Desk & Capacity):** Bed 55 is pre-approved by the doctor. Clicking "Finalize Discharge" frees Bed 55, updating the 74-bed matrix to Available (Green).
4. **Executive Telemetry:** Head Doctor dashboard computes live mortality vs the 15% threshold and cure rate vs 85%, alongside shift completion gauges.

Immediate Vercel Deployment Instructions

```
# Method 1: Instant Vercel Deployment via CLI cd C:\Users\kaulj\.gemini\antigravity\scratch\quarantine-facility-portal npx vercel # Method 2: Push to your GitHub and connect to Vercel git remote add origin https://github.com/[YOUR_USERNAME]/quarantine-facility-portal.git git push -u origin main # In vercel.com -> "Add New Project" -> Import repository -> Deploy
```